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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:24:42",
  "variant" : 1,
  "subjectName" : "mathematics",
  "categories" : [ "Probability", "Stereometry", "Planimetry" ],
  "questions" : [ {
    "question" : "What is the probability of flipping a fair coin three times and getting all heads?",
    "answers" : [ "1/4", "1/2", "1/6", "1/8" ],
    "correctIndex" : 3
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    "question" : "If events A and B are independent, what is P(A and B)?",
    "answers" : [ "P(A) + P(B) - P(A or B)", "P(A) * P(B)", "P(A) / P(B)", "P(A) + P(B)" ],
    "correctIndex" : 1
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    "question" : "What does Bayes' Theorem calculate?",
    "answers" : [ "Conditional probability", "Joint probability", "Marginal probability", "Discrete distributions" ],
    "correctIndex" : 0
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    "question" : "What is the sum of probabilities of all possible mutually exclusive outcomes in a sample space?",
    "answers" : [ "1", "100", "Infinite", "0" ],
    "correctIndex" : 0
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    "question" : "What is the volume formula for a perfect sphere with radius r?",
    "answers" : [ " $\pi * r^2$ ", " $2 * \pi * r$ ", " $4 * \pi * r^2$ ", " $(4/3) * \pi * r^3$ " ],
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    "question" : "What is the volume of a cone with base radius r and height h?",
    "answers" : [ " $2 * \pi * r * h$ ", " $(1/3) * \pi * r^2 * h$ ", " $\pi * r^2 * h$ ", " $(4/3) * \pi * r^2 * h$ " ],
    "correctIndex" : 1
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    "question" : "According to Brahmagupta's formula, what does it calculate?",
    "answers" : [ "Volume of a pyramid", "Perimeter of an ellipse", "Area of a cyclic quadrilateral", "Area of a scalene triangle" ],
    "correctIndex" : 2
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    "question" : "What is the length of the hypotenuse of a right-angled triangle with sides 5 and 12?",
    "answers" : [ "15", "17", "13", "14" ],
    "correctIndex" : 2
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"question" : "What is the sum of the interior angles of a regular hexagon?",
"answers" : [ "360 degrees", "900 degrees", "720 degrees", "540 degrees" ],
"correctIndex" : 2
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"question" : "Which theorem states that the square of the hypotenuse is equal to the sum of
squares of the other two sides?",
"answers" : [ "Euclid's Theorem", "Pythagorean Theorem", "Ptolemy's Theorem", "Thales's
Theorem" ],
"correctIndex" : 1
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